



International Business,
Science And Technology
UNIVERSITY
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ENGINEERING MATHEMATICS II REVISION QUESTIONS

QUESTION ONE

- (a) Find the orthogonal trajectories of $cx^2 - y^2 = 1$ (10 Marks)
- (b) A metal bar at a temperature of $100^\circ F$ is placed in a room at a constant temperature of $0^\circ F$, if after 20 minutes the temperature of the bar is $50^\circ F$. Find the time it takes the bar to reach the temperature of $25^\circ F$ and the temperature of the bar after 10 minutes (15 Marks)
- (c) Find the orthogonal trajectories of $Y^2 + (X - C)^2 = C^2$ (10 Marks)

QUESTION TWO

Solve the following differential equation

- (a) $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$ by letting $y = ux$ (10 Marks)
- (b) $\frac{dy}{dx} + 2y = 2e^x$ (5 Marks)
- (c) $\frac{dy}{dx} + y = \sin x$ (5 Marks)
- (d) $\frac{dy}{dx} - 2xy = x$ (5 Marks)
- (e) Use $y = ux$ to solve $\frac{dy}{dx} = \frac{y}{x+y}$ (6 Marks)
- (f) Use the method of variation of parameter to solve
- (i) $x \ln x \frac{dy}{dx} + y = \ln x$ (7 Marks)
- (ii) $X(X+1) \frac{dY}{dX} - (X+2) = X^3(X-3)$ (8 Marks)

QUESTION THREE

- (a) Use the method of variation of parameter to solve
- (i) $x^2 \frac{d^2y}{dx^2} - 4x \frac{dy}{dx} + 4y = x^2(x^2 + 1)$ for $x > 0$ (15 Marks)
- (ii) $y'' + 4y = \tan 2x$ (15 Marks)
- (iii) $\frac{d^2Y}{dX^2} + Y = \operatorname{CoSec} X$ (15 Marks)
- (iv) $\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + 5y = e^{-x} \sec(2x)$ for $x > 0$ (15 Marks)

(b) Solve the following differential equation

(i) $y'' - y = e^{2x}$ (7 Marks)

(ii) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 4y = 4x^2 - 2x$ (8 Marks)

(iii) $\frac{d^2y}{dx^2} + y = \sin x$ (10 Marks)

(iv) $\frac{d^2y}{dx^2} + y = \sin(2x)$ (7 Marks)

(v) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x}$ (8 Marks)

(c) Solve the following second order differential equations

(i) $X^2 \frac{d^2Y}{(dX)^2} - 2X \frac{dY}{dX} + 2Y = X^3 \cos x$ (15 Marks)

(ii) $Y'' + 3Y' + 2Y = \sin(e^x)$ (15 Marks)

QUESTION FOUR

Solve $X^3 \frac{d^3Y}{dX^3} - 4X^2 \frac{d^2Y}{dX^2} + 8X \frac{dY}{dX} - 8Y = 6X^3(X^2 + 1)^{-\frac{3}{2}}$ (20 Marks)

QUESTION FIVE

(a) Find f if $f''(x) = 4 - 6x - 40x^3$, $f(0) = 2$ and $f'(0) = 1$ (4 Marks)

(b) Evaluate the following integrals (4marks)

(a) $\int e^{-3x+1} dx$ (3 Marks)

(b) $\int \frac{x^2}{x^3+1} dx$ (4 Marks)

(c) $g(x) = \int_x^2 t^2 \sin t dt$ (4 Marks)

(c) Obtain the formula for $\int \cos^n x dx$ and hence integrate $\int \cos^4 x dx$ (10 Marks)

QUESTION FIVE

(a) Verify the divergence theorem for $A = 3xyz\mathbf{i} + z^2\mathbf{j} - 5yz\mathbf{k}$ taken over the region in the first octant bounded by $x = 2$ and $y^2 + z^2 = 9$ (15 Marks)

(b) Evaluate $\iint F ds$ for $F = (x^2 + y^2)\mathbf{i} - 2z\mathbf{j} + 2yz\mathbf{k}$, the surface S being part of the plane $2x + y + 2z = 6$ which lies in the first octant (10 Marks)

(c) Evaluate $\iint A \cdot ds$ for $A = 18z\mathbf{i} - 12z\mathbf{j} + 3yz\mathbf{k}$, the surface S being part of the plane $2x + 3y + 6z = 12$ which lies in the first octant (15 Marks)

(d) Evaluate $\iint F \cdot ds$ for $F = xyz\mathbf{i} - x^2z\mathbf{j} + (x+z)\mathbf{k}$, the surface S being part of the plane $2x + 2y + z = 6$ which lies in the first octant (20 Marks)

(e) Use Stoke's theorem to evaluate $\int (x+2y)dx + (x-z)dy + (y-z)dz$ where C is the boundary of the triangle with vertices at $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 6)$ normal to $3x + 2y + z = 6$ oriented in anticlockwise direction. (15 Marks)

(f) Use Stoke's theorem to verify $\iint \text{curl } A \cdot nds = \int A \cdot dr$ Given $A = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2zk\mathbf{k}$ where S is the upper half of a sphere $x^2 + y^2 + z^2 = 1$ (20 Marks)

(g) Verify the Green's theorem for $\int (2x - y^3)dx - xydy$ along $x^2 + y^2 = 1$ (20 Marks)

- (h) Evaluate $\iiint \phi dv$ for $\phi = 45x^2y$ and V denotes the region bounded by plane $4x + 2y + z = 8$ **(15 Marks)**
- (i) Evaluate $\int A \cdot dr$ for $A = (3x^2 + 6y)i - 14yzj + 20xz^2k$, from $(0, 0, 0)$ to $(1, 1, 1)$ along $x = t$, $y = t^2$ and $z = t^3$ **(15 Marks)**
- (j) Given $V = x^2zi - 2y^3z^2j + xy^2zk$ find
- (i) Divergence at point $(1, -1, 1)$ **(5 Marks)**
- (ii) Curl at point $(1, 1, 1)$ **(5 Marks)**
- (k) Given $F = 3ui + u^2j + (u + 2)k$ and $V = 2ui - 3uj + (u - 2)k$, evaluate $\int_0^2 F \times V du$ **(5 Marks)**

Prepared by:-

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